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Bonding method and apparatus of substrates

Abstract

A bonding method of substrates for bonding a first substrate and a second substrate. The method includes: placing the first substrate on a holding surface of a holder in a confined chamber; evacuating air from the confined chamber to vacuum the confined chamber to a predetermined low pressure value; placing the second substrate on a top of the first substrate; pressing to the second substrate onto the first substrate; and allowing an outside atmosphere air to slowly enter the confined chamber by a first flow coefficient; and allowing the outside atmosphere air to rapidly enter the confined chamber with a second flow coefficient, wherein the second flow coefficient is greater than the first flow coefficient.

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Background/Summary

BACKGROUND

Technical Field

(1) This disclosure relates to bonding substrates, and in particular, to a bonding method and bonding apparatus of substrates.

Related Art

(2) In the wafer process, an adhesive is applied to bonding two substrates before processing the substrates. Take the wafer thinning process as an illustration, the wafer to be thinned has insufficient stress strength and is prone to breakage. Therefore, before thinning process, the wafer is bonded to a carrier plate, so as to enhance the mechanical strength of the wafer. And then wafer is then subjected to thinning operations such as chemical-mechanical grinding.

(3) As described above, the wafer and the carrier plate are bonded through the adhesive. The adhesive must be sufficiently coated to the wafer and the carrier plate, and air between the wafer and the carrier plate has to be evacuated, so as to prevent air bubbles in the adhesive from affecting the bonding strength.

(4) The existing practice is to place the wafer and carrier plate in a confined chamber during the bonding process. When applying pressure to the two substrates, the confined chamber is simultaneously vacuumed to a low pressure. The air bubbles are attracted by the low pressure towards the edge of the substrates, and then the air between the wafer and the carrier plate is removed from the edge of the substrates.

(5) However, During the process of restoring the pressure in the confined chamber to atmospheric pressure (equilibrium with the outside atmospheric pressure), Air can re-enter between the wafer and the carrier plate and form new air bubbles affecting the bonding strength.

SUMMARY

- (6) In view of the above problem, this disclosure discloses a bonding method and apparatus for substrates, which is able to fully remove the air between the two substrates being bonded.
- (7) This disclosure discloses a bonding method of substrates for bonding a first substrate and a second substrate. The method comprises: placing the first substrate on a holder in a confined chamber; evacuating air from the confined chamber to vacuum the confined chamber to a predetermined low pressure value; placing the second substrate on a top of the first substrate; pressing to the second substrate onto the first substrate; and allowing an outside atmosphere air to slowly enter the confined chamber by a first flow coefficient; and allowing the outside atmosphere air to rapidly enter the confined chamber with a second flow coefficient, wherein the second flow coefficient is greater than the first flow coefficient.
- (8) In one or more embodiments, the step of providing the confined chamber comprises: providing a vacuum container, wherein the confined chamber is defined within the confined chamber.
- (9) In one or more embodiments, the first substrate is placed on a holding surface of the holder at first, wherein the holder is disposed on a bottom of the vacuum container with the holding surface facing a top of the vacuum container.
- (10) In one or more embodiments, the step of evacuating air from the confined chamber comprises: connecting a vacuum pump to the confined chamber by an air evacuation pipe and evacuating air by the vacuum pump.
- (11) In one or more embodiments, the step of pressing the second substrate comprises: disposing a bonding head to the top of the vacuum container in a movable manner, allowing the bonding head to move toward the holder to contact and press the second substrate.
- (12) In one or more embodiments, the step of allowing the outside atmosphere air to enter the confined chamber comprises: connecting a two-stage valve assembly to the confined chamber via an air inlet pipe; wherein a flow coefficient of the two-stage valve assembly is configured to be selectively switched into a first flow coefficient and a second flow coefficient, and the second flow coefficient is greater than the first flow coefficient.
- (13) In one or more embodiments, the two-stage valve assembly includes: a first shut-off valve and a throttle valve connected in series with the air inlet pipe, wherein the first shut-off valve is disposed between the air inlet pipe and the throttle valve, and an equivalent flow coefficient of the first shut-off valve and the throttle valve connected in series equals the first flow coefficient; and a second shut-off valve with two ends respectively connected to an inlet end of the first shut-off valve and an outlet end of the throttle valve, and when the first shut-off valve and the second shut-off valve simultaneously open, an equivalent flow coefficient equals the second flow coefficient.
- (14) In one or more embodiments, when allowing the outside atmosphere air to enter the confined chamber by the first flow coefficient, the first shut-off valve is opened to allow the outside atmosphere air to enter the confined chamber via the throttle valve and the first shut-off valve.
- (15) In one or more embodiments, when allowing the outside atmosphere air to enter the confined chamber by the second flow coefficient, the second shut-off valve is opened to allow the outside atmosphere air to enter the confined chamber simultaneously via the first shut-off valve, the second shut-off valve, and the throttle valve.
- (16) In one or more embodiments, the bonding method further comprises applying an adhesive to the first substrate or the second substrate, to allow the adhesive to be disposed between the first substrate and the second substrate.
- (17) This disclosure further discloses a bonding apparatus of substrates for bonding a first substrate and a second substrate, bonding apparatus comprises: a vacuum container having a confined chamber defined therein, wherein the confined chamber is configured to dispose the first substrate and the second substrate therein; a bonding head disposed to the vacuum container and located in the confined chamber, wherein the bonding head is configured to press the second substrate onto the first substrate; an air evacuating pipe configured to connect the confined chamber to a vacuum pump, to evacuate air from the confined chamber; an air inlet pipe connected to the confined

chamber; and a two-stage valve assembly connected to the confined chamber via the air inlet pipe, wherein the two-stage valve assembly is configured to allow an outside atmosphere air to enter the confined chamber to raise a pressure in the confined chamber; wherein a flow coefficient of the two-stage valve assembly is configured to be selectively switched into a first flow coefficient and a second flow coefficient, and the second flow coefficient is greater than the first flow coefficient.

(18) In one or more embodiments, when raising the pressure in the confined chamber, the two-stage valve assembly opens by the first flow coefficient for a predetermined time period, and then opens by the second flow coefficient.

(19) In one or more embodiments, the two-stage valve assembly includes: a first shut-off valve and a throttle valve connected in series with the air inlet pipe, and wherein the first shut-off valve is disposed between the air inlet pipe and the throttle valve, and an equivalent flow coefficient of the first shut-off valve and the throttle valve connected in series equals the first flow coefficient; and a second shut-off valve with two ends respectively connected to an inlet end of the first shut-off valve and an outlet end of the throttle valve, and when the first shut-off valve and the second shut-off valve simultaneously open, an equivalent flow coefficient equals the second flow coefficient.

(20) In one or more embodiments, when raising the pressure in the confined chamber, the first shut-off valve opens for a predetermined time period at first, and the second shut-off valve opens.

(21) In one or more embodiments, the vacuum container includes an upper container and a lower container, and the upper container is combined with the lower container to define the confined chamber.

(22) In one or more embodiments, the bonding apparatus further comprises a holder, disposed on the lower container and located within the confined chamber, wherein a holding surface of the holder faces the upper container for holding the first substrate.

(23) In one or more embodiments, the bonding head is disposed to the upper container in a movable manner.

(24) In one or more embodiments, the bonding apparatus further comprises a linear actuator disposed outside the upper container, wherein an actuator rod of the linear actuator passes through the upper container and connected to the bonding head, allowing the bonding head to move toward the holder.

(25) Through the bonding method and apparatus of this disclosure, the air between the bonded substrates can be fully eliminated, and at the same time, air is prevented from re-entering between the two substrates during the pressure recovery process, effectively improving the bonding yield.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) This disclosure will become more fully understood from the detailed description given herein below for illustration only, and thus not limitative of this disclosure, wherein:
- (2) FIG. 1 is a cross-sectional view of a bonding apparatus according to an embodiment of this disclosure.
- (3) FIG. 2 is another cross-sectional view of the bonding apparatus according to the embodiment of this disclosure.
- (4) FIG. 3 is yet another cross-sectional view of the bonding apparatus according to the embodiment of this disclosure.
- (5) FIG. 4 is a flowchart of a bonding method according to the embodiment of this disclosure.
- (6) FIG. 5 is a cross-sectional view of the bonding apparatus showing part of the steps of the bonding method.
- (7) FIG. 6 and FIG. 7 are partial cross-sectional views of the bonding apparatus, showing part of the steps of the bonding method.

DETAILED DESCRIPTION

(8) As shown in FIG. 1, FIG. 2 and FIG. 3, a bonding apparatus **100** for substrates according to an embodiment of this disclosure is shown, which is configured to perform a binding method to bond a first substrate **S1** and a second substrate **S2**. The first substrate **S1** and the second substrate **S2** may be, but not limited to, a wafer to be thinned and a carrier plate for strengthening mechanical strength. It should be noted that The dimensional proportions of the first substrate **S1** and the second substrate **S2** drawn in the drawings is for the purpose of making this disclosure to be easily understood, it does not represent the actual size ratio of the first substrate **S1** and the second substrate **S2**.

(9) As shown in FIG. 1, FIG. 2 and FIG. 3, The bonding apparatus **100** includes a lower container **110**, an upper container **120**, a holder **130**, a bonding head **140**, an air evacuating pipe **150**, an air inlet pipe **160**, and a two-stage valve assembly **170**.

(10) As shown in FIG. 1, FIG. 2 and FIG. 3, the upper container **110** and the lower container **120** are combined to a vacuum container **100a**, and a confined chamber **100b** is defined within the vacuum container **100a**. The lower container **110** forms the bottom of the vacuum container **100a**, and the upper container **120** forms the top of the vacuum container **100a**. The holder **130** is disposed on the bottom of the vacuum container **100a**, that is the holder **130** is disposed on the lower container, and the holder **130** is located within the confined chamber **100b**. A holding surface **132** of the holder **130** faces the top of the vacuum container **100a**, that is the holding surface **132** faces the upper container **120**. The holding surface **132** of the holder **130** is configured to hold the first substrate **S1**. Specifically, the lower container **110** may fixed to a base, and the upper container **120** is connected to a lifting device. Raising or lowering the upper container **120** by the lifting device, the upper container **120** and the lower container **110** are allowed to separate from each other to open the vacuum container **100a**, or the upper container **120** is allowed to be combined with the lower container **110** to form the vacuum container **100a** and the confined chamber **100b**.

(11) As shown in FIG. 1, FIG. 2 and FIG. 3, the bonding head **140** is disposed to the upper container **120** of the vacuum container **100a** in a movable manner, such that the bonding head **140** is disposed to the top of the vacuum container **100a**, and the bonding head **140** is located within the confined chamber **100b**. The bonding head **140** is correspondingly disposed above the holder **130**, and is configured to press the second substrate **S2** onto the first substrate **S1**, such that an adhesive **G** between the two substrate is able to bond the second substrate **S2** to the first substrate **S1**, and air between the first substrate **S1** and the second substrate **S2** is discharged.

(12) As shown in FIG. 1, FIG. 2, and FIG. 3, specifically, the bonding apparatus **100** further includes a linear actuator **180** disposed outside the upper container **120**. An actuator rod **182** of the linear actuator **180** passes through the upper container **120** and connected to the bonding head **140**, to allow the bonding head **140** to move toward the holder **130** or away from the holder **130**. An airtight shaft seal is provided at the joint of the actuator rod **182** and the upper container **120** to maintain the airtightness of the confined chamber **100b**.

(13) As shown in FIG. 1, FIG. 2 and FIG. 3, the air evacuating pipe **150** is disposed to the vacuum container **100a** for connecting the confined chamber **100b** to a vacuum pump **190**. The vacuum pump **190** is configured to evacuate air from the confined chamber **100b**, so as to vacuum the confined chamber **100b** to a predetermined low-pressure value. Specifically, the air evacuating pipe **150** may be disposed to the lower container **110** or disposed to the upper container **120**.

(14) As shown in FIG. 1, FIG. 2 and FIG. 3, the air inlet pipe **160** is disposed to the vacuum container **100a** for connecting the confined chamber **100b** to exterior. Similarly, the air inlet pipe **160** may be disposed to the lower container **110** or disposed to the upper container **120**.

(15) As shown in FIG. 1, FIG. 2 and FIG. 3, the two-stage valve assembly **170** is connected to the air inlet pipe **160**, so as to be connected to the confined chamber **100b** via the air inlet pipe **160**. A flow coefficient **Cv** of the two-stage valve assembly is configured to be selectively switched into a first flow coefficient **Cv1** and a second flow coefficient **Cv2**, and the second flow coefficient **Cv2** is

greater than the first flow coefficient **Cv1**. The two-stage valve assembly **170** is configured to allow an outside atmosphere air to enter the confined chamber **100b** to raise a pressure in the confined chamber **100b**. When the two-stage valve assembly **170** opens and is switched to have the first flow coefficient **Cv1**, a pressure in the confined chamber **100b** slowly raises. When the two-stage valve assembly **170** opens and is switched to have the second flow coefficient **Cv2**, the pressure in the confined chamber **100b** rapidly raises to equalize with the outside atmosphere.

(16) As shown in FIG. 1, FIG. 2 and FIG. 3, the two-stage valve assembly **170** includes a first shut-off valve **171**, a throttle valve **173**, and a second shut-off valve **172**. The first shut-off valve **171** and a throttle valve **173** are connected in series with the air inlet pipe **160**, and the first shut-off valve **171** is disposed between the air inlet pipe **160** and the throttle valve **173**. Two ends of the second shut-off valve **172** is respectively connected to an inlet end of the first shut-off valve **171** and an outlet end of the throttle valve **173**, that is the second shut-off valve **172** the inlet end of the first shut-off valve **171** to the outlet end of the throttle valve **173**. The first shut-off valve **171** and the second shut-off valve **172** may be identical shut-off valves having the same fluid coefficient **Cv**. the throttle valve **173** is configured to regulate the flow rate of air through the second shut-off valve **172**. The throttle valve **173** may be pre-regulated to such an extent that the first shut-off valve **171** in series with the throttle valve **173** provides a fluid coefficient **Cv** equal to the desired first fluid coefficient **Cv1**. And when the first shut-off valve **171** and the second shut-off valve **172** are opened simultaneously, the two-stage valve assembly **170** has an equivalent second fluid coefficient **Cv2**. And the second flow coefficient **Cv2** is greater than the first flow coefficient **Cv1**.

(17) Referring to FIG. 4, a bonding method for substrates of this disclosure is shown.

(18) As shown in FIG. 1, FIG. 2 and FIG. 4, first, the method is to dispose the first substrate **S1** in the confined chamber **100b**, wherein the first substrate **S1** is placed on a holding surface **132** of the holder **130** at first, and the holder is disposed on a bottom of the vacuum container **100a**, as shown in step **S110**.

(19) The first substrate **S1** may be placed in the confined chamber **100b** that is half-open when the confined chamber **100b** is not fully closed, i.e., when the upper container **120** is not yet combined with the lower container **110**, and is disposed on the holding surface **132** of the bearing table **130**. The first substrate **S1** and the second substrate **S2** may also be placed in the semi-open closed chamber **100b**, and the first substrate **S1** and the second substrate **S2** may be grasped by a fixture or transfer tool (not shown). Next, the method is to combine the upper container **120** and the lower container **110** to form the confined chamber **100b**. The first substrate **S1** is then placed on the holding surface **132** of the holder **130** by means of clamps and transfer devices. The method may also be to place the first substrate **S1** on the holding surface **132** of the holder **130** at first when the confined chamber **100b** is half-open.

(20) As shown in FIG. 2 and FIG. 4, have the confined chamber **100b** to be airtight, such as by combining the upper container **120** with the lower container **110**, and evacuate air from the confined chamber **100b** by using a vacuum pump **190**, so as to vacuum the confined chamber **100b** to a predetermined low-pressure value, such as a low pressure close to a vacuum, as shown in step **S120**.

(21) As shown in FIG. 3 and FIG. 4, place the second substrate **S1** onto the first substrate **S1**, as shown in step **S130**. One of the first substrate **S1** and the second substrate **S2** may be coated with an adhesive **G** in advance, to have the adhesive **G** located between the first substrate **S1** and the second substrate **S2**. The adhesive **G** may be coated on the upper surface of the first substrate **S1** or coated on the lower surface of the second substrate **S2**.

(22) In order to avoid the formation of air bubbles over a large area directly when the second substrate **S2** is placed on top of the first substrate **S1**, the second substrate **S2** may be tilted to be placed on the first substrate **S1**, the second substrate **S2** is then gradually affixed to the first substrate **S1** so that the large air bubbles can be slowly discharged.

(23) As shown in FIG. 3, FIG. 4 and FIG. 5, In a state where the pressure in the confined chamber

100a is low pressure, activate the linear actuator **180** to move the bonding head **140** toward the holder **130**, such that the bonding head **130** contacts the second substrate **S2** and then presses the second substrate **S2** onto the first substrate **S1** to form a bonded substrate, as shown in step **S140**.

(24) As shown in FIG. 5, at the time, attracted by the low pressure of the confined chamber **100b**, air bubbles remaining in the adhesive **G**, i.e., air bubbles between the first substrate **S1** and the second substrate **S2**, will continue to move toward the edges of the bonded substrates, allowing the air between the first substrate **S1** and the second substrate **S2** to be discharged. Therefore, the first substrate **S1** and the second substrate **S2** can be well bonded by the adhesive **G**, avoiding discontinuities in the bonding of the adhesive **G**.

(25) As shown in FIG. 4 and FIG. 6, then, open the first shut-off valve **171** to allow the outside atmosphere air to slowly enter the confined chamber **100b** by the first flow coefficient through the throttle valve **173** and the first shut-off valve **171** of the two-stage valve assembly **170**, such that the pressure in the confined chamber **100b** raises slowly, as shown in step **S150**.

(26) At the time, the confined chamber **100b** remains relatively low-pressure state. Therefore, if there are still air bubbles **B** between the first substrate **S1** and the second substrate **S2**, the air bubbles **B** can still be discharged by low pressure attraction. The slow rise in pressure also prevents air from re-entering between the first substrate **S1** and the second substrate **S2** and creating new air bubbles **B**.

(27) As shown in FIG. 4 and FIG. 7, After a certain pressurization time, the pressure inside the confined chamber **100b** reaches a certain ratio to the outside atmospheric pressure. At this time, open the second opening valve **172** so that the outside atmospheric air also enters the confined chamber **100b** through the second opening valve **172** simultaneously. At the time, the outside atmosphere air rapidly enters the confined chamber **100b** through the first shut-off valve **171**, the second shut-off valve **172**, and the shut-off valve **173** of the two-stage valve assembly **170**, such that the pressure in the confined chamber **100b** raises rapidly to equalize with the outside atmospheric pressure, as shown in step **S160**.

(28) Finally, the linear actuator **180** drives the bonding head **140** to leave the second substrate **S2** and the upper container **120** is separated from the lower container **110** to open the vacuum chamber **100a**, so as to take out the bonded substrates composed of the first substrate **S1** bonded to the second substrate **S2**, and to complete the operation of the substrates bonding.

(29) Through the bonding method and apparatus of this disclosure, the air between the bonded substrates can be fully eliminated, and at the same time, air is prevented from re-entering between the two substrates during the pressure recovery process, effectively improving the bonding yield.

Claims

1. A bonding method of substrates for bonding a first substrate and a second substrate, comprising: placing the first substrate on a holder in a confined chamber; evacuating air from the confined chamber to vacuum the confined chamber to a predetermined low-pressure value; placing the second substrate on a top of the first substrate; pressing the second substrate onto the first substrate; allowing an outside atmosphere air to enter the confined chamber by a first flow coefficient; and allowing the outside atmosphere air to enter the confined chamber by a second flow coefficient; wherein the second flow coefficient is greater than the first flow coefficient.
2. The bonding method as claimed in claim 1, wherein the first substrate is placed on a holding surface of the holder at first, wherein the holder is disposed on a bottom of a vacuum container with the holding surface facing a top of the vacuum container.
3. The bonding method as claimed in claim 1, wherein the step of evacuating air from the confined chamber comprises: connecting a vacuum pump to the confined chamber by an air evacuation pipe and evacuating air by the vacuum pump.
4. The bonding method as claimed in claim 1, wherein the step of pressing the second substrate

- comprises: disposing a bonding head to the top of a vacuum container in a movable manner, allowing the bonding head to move toward the holder to contact and press the second substrate.
5. The bonding method as claimed in claim 1, wherein the step of allowing the outside atmosphere air to enter the confined chamber by the first flow coefficient or the step of allowing the outside atmosphere air to enter the confined chamber by the second flow coefficient comprises: connecting a two-stage valve assembly to the confined chamber via an air inlet pipe; wherein a flow coefficient of the two-stage valve assembly is configured to be selectively switched into the first flow coefficient and the second flow coefficient.
6. The bonding method as claimed in claim 5, wherein the two-stage valve assembly includes: a first shut-off valve and a throttle valve connected in series with the air inlet pipe, wherein the first shut-off valve is disposed between the air inlet pipe and the throttle valve, and a first equivalent flow coefficient of the first shut-off valve and the throttle valve connected in series equals the first flow coefficient; and a second shut-off valve with two ends respectively connected to an inlet end of the first shut-off valve and an outlet end of the throttle valve, and when the first shut-off valve and the second shut-off valve simultaneously open, a second equivalent flow coefficient equals the second flow coefficient.
7. The bonding method as claimed in claim 6, wherein when allowing the outside atmosphere air to enter the confined chamber by the first flow coefficient, the first shut-off valve is opened to allow the outside atmosphere air to enter the confined chamber via the throttle valve and the first shut-off valve.
8. The bonding method as claimed in claim 6, when allowing the outside atmosphere air to enter the confined chamber by the second flow coefficient, the second shut-off valve is opened to allow the outside atmosphere air to enter the confined chamber simultaneously via the first shut-off valve, the second shut-off valve, and the throttle valve.
9. The bonding method as claimed in claim 6, further comprising: applying an adhesive to the first substrate or the second substrate, to allow the adhesive to be disposed between the first substrate and the second substrate.
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